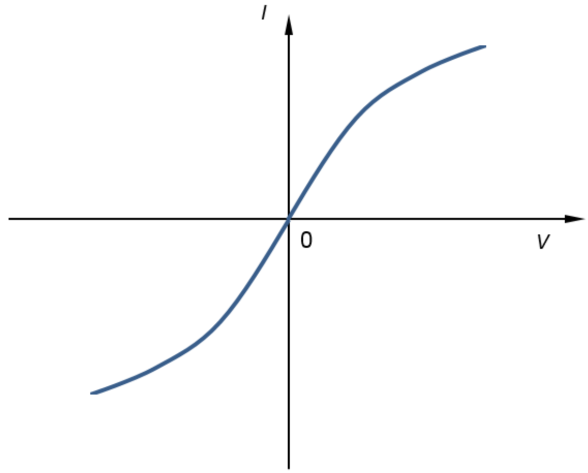
$= 5.00 \times 10^{[illegible]}$ s

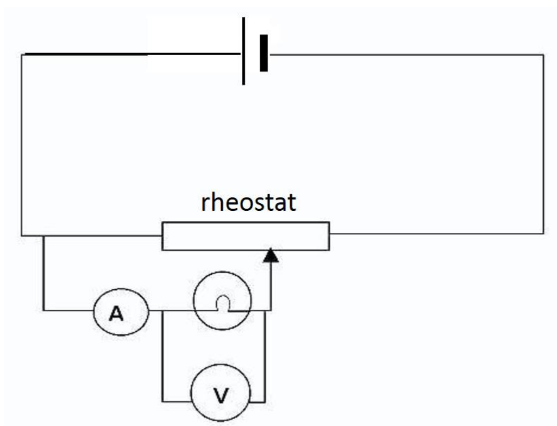
9(a)(i) The ohm is the unit of resistance such that 1 Ω is the resistance where the potential difference between 2 points across a conductor is 1 V produces a current of 1 A flowing through it.

(ii)




(iii) Resistance R is the ratio of the p.d V to the current I at that particular p.d on the graph.

(iv)




b(i) Total resistance of the 4 parallel resistors = $(1/2R + 1/2R)^{-1} = R$.
$I = E/(R + r)$.

(ii) Power in the 4 parallel resistors = $I^2 R$, where $I = E/(R+r)$
Power in the battery = $IE$

$$\frac{power\ in\ external\ resistors}{power\ by\ batter} = \frac{I^2 R}{IE}$$

$$= \frac{I\ R}{E}$$

$$= \frac{E\ R}{R + r}\frac{1}{E}$$

$$= \frac{R}{R + r}$$

(iii) Having 4 resistors in parallel, each resistor only dissipates ¼ of the power compared to a single resistor.
Hence 4 resistors in parallel can have a higher power rating than just a single resistor.

(c)(i) When light shone on LDR, resistance of LDR is 420 Ω.
p.d across LDR = $\frac{420}{420+8200}$ × 6.0 = 0.29 $V$

Hence potential at P, $V_P$ = 0.29 – 2.0 = -1.71 V

(ii) When LDR in the dark, resistance of LDR = 134 kΩ.
p.d across LDR = $\frac{134\ \text{k}\Omega}{134\ \text{k}\Omega+8.2\ \text{k}\Omega}$ × 6.0 = 5.65 $V$

Hence potential at P, $V_P$ = 5.65 – 2.0 = 3.65 V.

(iii) The alarm is placed in parallel with the LDR.
When the laser beam is shining on the LDR, the resistance of the LDR will be low. By potential divider principle, the p.d. across the LDR and alarm will be low. The alarm will remain turn off.
When the burglar blocked the laser beam, the resistance of the LDR will be high. Hence the p.d across the LDR and alarm will be high. This will turn on the alarm.

(d)(i) There will be no deflection in the galvanometer when the p.d across JY = 1.5 V.
Let the balance length (between J and Y) be *L*. Then

$$V_{JY} = \frac{L \times 1.50}{(1.50+0.50)} \times 4.0 = 1.5$$

$L$ = 0.500 m.

Hence distance from X = 1.00 – $L_{JY}$ = 0.50 m.

(ii) When 1.50 Ω connected with the 1.5 V cell, the terminal p.d across the 1.50 Ω is

$$V = \frac{1.50}{1.50+0.50} \times 1.5 = 1.13\,V$$

There will null deflection when $V_{JY}$ = 1.13 V. Then

$$V_{JY} = \frac{L' \times 1.50}{(1.50+0.50)} \times 4.0 = 1.13$$

$L'$ = 0.377 m.

Hence distance from X = 1.00 – 0.377 = 0.62 m.